ON THE REPRESENTATIONS OF FINITE DISTRIBUTIVE LATTICES

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Abstract. A simple but elegant result of Rival states that every sublattice L of a finite distributive lattice $\mathcal{P}$ can be constructed from $\mathcal{P}$ by removing a particular family $\mathcal{I}_L$ of its irreducible intervals.

Applying this in the case that $\mathcal{P}$ is a product of a finite set $\mathcal{C}$ of chains, we get a one-to-one correspondence $L \mapsto \mathcal{D}_{\mathcal{P}}(L)$ between the sublattices of $\mathcal{P}$ and the preorders spanned by a canonical sublattice $\mathcal{C}^{\infty}$ of $\mathcal{P}$.

We then show that L is a tight sublattice of the product of chains $\mathcal{P}$ if and only if $\mathcal{D}_{\mathcal{P}}(L)$ is asymmetric. This yields a one-to-one correspondence between the tight sublattices of $\mathcal{P}$ and the posets spanned by its poset $J(\mathcal{P})$ of non-zero join-irreducible elements.

With this we recover and extend, among other classical results, the correspondence derived from results of Birkhoff and Dilworth, between the tight embeddings of a finite distributive lattice L into products of chains, and the chain decompositions of its poset $J(L)$ of non-zero join-irreducible elements.

1. Introduction

All lattices considered in this paper are finite and distributive. For very basic notation, definitions, and concepts we refer the reader to [4]; many basic definitions are also given at the start of Section 2.

Classical results of Birkhoff and Dilworth, which we review in more detail in Section 2, show that any finite distributive lattice L can be embedded, as a sublattice, into a product of chains. They further yield a one-to-one correspondence between the tight such embeddings (those that preserve covers, and the the zero and unit elements), and chain decompositions of the poset $J(L)$ of non-zero irreducible elements of L.

In this paper, we consider the following question:

> What happens when we reverse our point of view, and ask about the various sublattices of a given products of chains $\mathcal{P}$ rather than the various embeddings of a particular lattice L into products of chains?

Starting with the product $\mathcal{P} = \prod_{C \in \mathcal{C}} C$ of a set $\mathcal{C}$ of chains, $J(\mathcal{P})$ is the parallel sum of the chains we get from the chains of $\mathcal{C}$ by removing their minimum elements. One main idea is that, for a tight sublattice L of $\mathcal{P}$, $J(L)$ is isomorphic to an extension of $J(\mathcal{P})$. See Figure 1 for an example. If we consider only tight

2010 *Mathematics Subject Classification.* 06D05.

Key words and phrases. Finite distributive lattice, Representation, Embedding, Product of chains.

Supported by the National Research Foundation (NRF) of Korea (2014-06060000) Posted on `arXiv:1412.0011 [math.CO]`.

ver. 12:41:44 2016/07/08 arXiv:1412.0011 [math.CO]

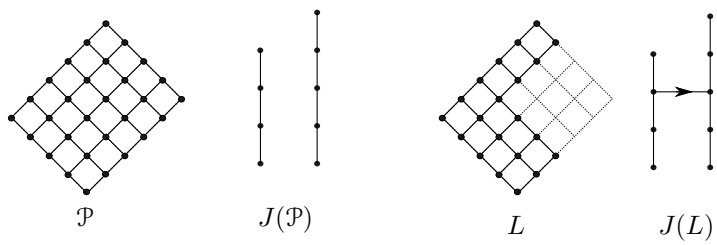

FIGURE 1. Product of chains $\mathcal{P}$, sublattice L, and their posets $J(\mathcal{P})$ and $J(L)$

sublattices of $\mathcal{P}$, this is the whole picture (Corollary 5.2): $L \mapsto J(L)$ gives a one-to-one correspondence of the tight sublattices of $\mathcal{P}$ and the spanned poset extensions of $J(\mathcal{P})$– extensions with the same set of elements as $J(\mathcal{P})$. But what if we consider non-tight embeddings / sublattices?

As every lattice has a tight embedding into products of chains, non-tight embeddings are seldom considered. However, in the paper [8], out of which this paper grew, we found it useful to consider non-tight embeddings, or rather non-tight sublattices of products of chains. We needed a characterisation of **all** sublattices of a product of chains.

Using a result of Rival [7] which characterises the sublattices of $\mathcal{P}$, we find a correspondence $L \mapsto D_{\mathcal{P}}(L)$ between the sublattices L of $\mathcal{P}$ and *preorders* (reflexive transitive relations) $D_{\mathcal{P}}(L)$ extending $J(\mathcal{P})$. To do this in full generality it is necessary to replace $J(\mathcal{P})$ with $\mathcal{C}^{\infty}$, which we get from $J(\mathcal{P})$ by appending a zero and a unit. Our main result yields Corollary 4.2, a one-to-one correspondence between sublattices of $\mathcal{P}$ and spanned preorder extensions of $\mathcal{C}^{\infty}$.

We then characterise the tight sublattices L of $\mathcal{P}$ as those for which $D_{\mathcal{P}}(L)$ is a poset and so, after restricting back to the elements of $J(\mathcal{P})$, find that in this case $D_{\mathcal{P}}(L)$ is $J(L)$. We further characterise the so-called $\{\mathbf{0}, \boldsymbol{\infty}\}$ and semi-direct sublattices of $\mathcal{P}$ in terms of properties of $D_{\mathcal{P}}$, and restricting our results to $\{\mathbf{0}, \boldsymbol{\infty}\}$-sublattices, get, in Corollary 5.2 a one-to-one correspondence between the $\{\mathbf{0}, \boldsymbol{\infty}\}$-sublattices of $\mathcal{P}$ and the spanned preorder extensions of $J(\mathcal{P})$.

Returning to the point of view of embeddings of a given lattice, we extend the classical correspondence between the tight embeddings of a lattice L into products of chains and the chain decompositions of $J(L)$. We get, in Theorem 6.5, a one-to-one correspondence between embeddings of L into products of chains, and surjective homomorphisms of 'pointed unions of chains' to $J^{\infty}(L)$.

In [5], Koh used a clever construction to show that any distributive lattice L can be represented as the lattice $\mathcal{A}_{\mathrm{m}}(P)$ of maximal antichains of some poset P. We finish off by showing that his construction arises naturally from our point of view, and extend it to determine, in Corollary 7.14, **all** posets P such that $L \cong \mathcal{A}_{\mathrm{m}}(P)$ for a given L.

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2. Notation and Background

A lattice embedding $e : L \to L'$ is *a* $\{\mathbf{0}, \boldsymbol{\infty}\}$*-embedding* if it preserves the zero and unit elements: $e(\mathbf{0}_L) = e(\boldsymbol{\infty}_{L'})$ and $e(\boldsymbol{\infty}_L) = \boldsymbol{\infty}_{L'}$. It is *tight* if it is a $\{\mathbf{0}, \boldsymbol{\infty}\}$-embedding and preserves covers: $x \prec y$ implies that $e(x) \prec e(y)$.

An element x of a lattice L is *join-irreducible* if $x = a \vee b$ implies that $x = a$ or $x = b$. Meet-irreducible elements are defined analogously. The set of non-zero join-irreducible elements of L is denoted $J(L)$. It induces a subposet of L which is also denoted by $J(L)$.

For a subset S of a lattice L, we let $\bigvee S = \bigvee_{x \in S} x$ be the join of the elements of S. We often write $\bigvee_L S$ to specify that the join takes place in L.

A subset S of a poset is a *downset* or *ideal* if $x \in S$ and $y \leq x$ implies $y \in S$. The minimum downset containing an element x is denoted $\operatorname{id} x$.

A *chain* C of length n in a poset P is subposet isomorphic to the linear order Z_n on the n elements $\{0, 1, \ldots, n-1\}$. A *chain decomposition* of a poset P is a partition of its elements into a family $\mathcal{C}$ of chains $C_1, \ldots, C_d$. For a family $\mathcal{C} = \{C_1, \ldots, C_d\}$ of disjoint chains, the product $\prod \mathcal{C} := \prod_{i=1}^{d} C_i$ consists of all d-tuples $x = (x_1, \ldots, x_d)$ where $x_i \in C_i$ for each $i \in [d]$. It is ordered by $x \leq y$ if $x_i \leq y_i$ for each i.

2.1. Embedding through Downsets.

In [1], Birkhoff showed that $L \cong \mathcal{D}(J(L))$ where $\mathcal{D}(P)$, for a poset P is the family of downsets of P ordered by inclusion. Specifically, he showed the following.

Theorem 2.1. [1] *The map* $\mathbf{S} : x \mapsto \operatorname{id} x \cap J(L)$ *is an isomorphism of* L *to* $\mathcal{D}(J(L))$*. Its inverse is* $S \mapsto \bigvee_L S$*.*

One can easily show that $P \cong J(\mathcal{D}(P))$ by observing that a downset in $\mathcal{D}(P)$ is join-irreducible if and only if it has a unique maximal element. Thus Theorem 2.1 gives the following, which completes a one-to-one correspondence between finite posets and finite distributive lattices.

Corollary 2.2. *If* $\mathcal{D}(P) \cong \mathcal{D}(P')$ *then* $P \cong P'$*.*

For a chain decomposition $\mathcal{C}$ of a poset let $\mathcal{C}_0$ be the family of chains we get from the chains in $\mathcal{C}$ by adding a new minimum element to each. In [2], Dilworth proved the following embedding theorem.

Theorem 2.3. [2] *For any chain decomposition* $\mathcal{C}$ *of a poset* P *the map* $S \mapsto \bigvee_{\mathcal{P}} S$ *is an embedding of* $\mathcal{D}(P)$ *into* $\mathcal{P} = \prod \mathcal{C}_0$*.*

With Theorem 2.1, this immediately gives the following.

Corollary 2.4. *For any decomposition* $\mathcal{C}$ *of* $J(L)$ *into chains, the map*

$$e_{\mathcal{C}} : L \to \prod \mathcal{C}_0 : x \mapsto \bigvee_{\mathcal{P}} \mathbf{S}(x)$$

is an embedding of L *into* $\mathcal{P} = \prod \mathcal{C}_0$*.*

In [6], Larson makes explicit a converse to Corollary 2.4, showing essentially the following.

Theorem 2.5. [6] *The embeddings* $e_{\mathcal{C}}$ *of Corollary 2.4 are tight, and for every tight embedding* e *there is a chain decomposition* $\mathcal{C}$ *of* $J(L)$ *such that* $e = e_{\mathcal{C}}$*.*

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It is a trivial observation that different chain decompositions of $J(L)$ yield different embeddings of L into products of chains, so with Corollary 2.4, this yields the following correspondence.

Corollary 2.6. *There is a one-to-one correspondence between the chain decompositions of $J(L)$ and the tight embeddings of L into products of chains.*

2.2. **Embeddings through antichains.** The correspondence taking a downset of a poset P to its subset of maximal elements gives one-to-one correspondence between the downsets $\mathcal{D}(P)$ of P and the antichains $\mathcal{A}(P)$ of P. It is not hard to see that by defining the following ordering on $\mathcal{A}(P)$, the correspondence can be extended to a lattice isomorphism: for antichains $I, I' \in \mathcal{A}(P)$, set $I \leq I'$ if for all $a \in I$ there is some $a' \in I'$ such that $a \leq a'$. In [3], Dilworth showed the following, which is considerably harder.

Theorem 2.7. [3] *The poset $\mathcal{A}_{\mathrm{m}}(P) \leq \mathcal{A}(P)$ of all maximum sized antichains of a poset P is a distributive lattice.*

In [5], Koh showed a converse: every finite distributive lattice L is $\mathcal{A}_{\mathrm{m}}(P)$ for some lattice P. In doing so he showed something stronger. Koh defines a construction $(J(L), \mathcal{C}) \mapsto P_{\mathcal{C}}$ which yields a poset $P_{\mathcal{C}}$ for every chain decomposition $\mathcal{C}$ of the poset $J(L)$. He then showed the following.

Theorem 2.8. [5] *For every finite distributive lattice L, and every chain decomposition $\mathcal{C}$ of $J(L)$, $L \cong \mathcal{A}_{\mathrm{m}}(P_{\mathcal{C}})$.*

2.3. **Rival's theorem.** An *irreducible interval* of a lattice L is the set $[\alpha, \beta] = \{x \in L \mid \alpha \leq x \leq \beta\}$ for any join-irreducible element α of L and any meet-irreducible element β. For a set $\mathcal{I}$ of irreducible intervals of L we let $\bigcup \mathcal{I} = \bigcup_{I \in \mathcal{I}} I$. The set $\mathcal{I}$ is *closed* if $I \subseteq \bigcup \mathcal{I}$ implies $I \in \mathcal{I}$. The key to our results is the following theorem of Rival.

Theorem 2.9. [7] *If L is a sublattice of a lattice L' of finite length, then*

$$L = L' \setminus \bigcup \mathcal{I}$$

for some (closed) family $\mathcal{I}$ of irreducible intervals of L'.

2.4. **Non-trivial Downsets.** For a poset P let $P_{\mathbf{0}}$ be the poset we get from P by adding a new minimum element $\mathbf{0}$ and let $P^{\boldsymbol{\infty}}$ be the poset we get from $P_{\mathbf{0}}$ by adding a new maximum element $\boldsymbol{\infty}$. A downset of a poset P is *non-trivial* if it is a proper non-empty subset of P. The following has been observed in several places, (see [6].)

Fact 2.10. *Where $\mathcal{D}_{\mathrm{ne}}(P)$ is the poset of* non-empty *downsets of a poset P, $\mathcal{D}(P) \cong \mathcal{D}_{\mathrm{ne}}(P_{\mathbf{0}})$. Where $\mathcal{D}_{\mathrm{nt}}(P)$ is the poset of* non-trivial *downsets of P, $\mathcal{D}(P) \cong \mathcal{D}_{\mathrm{nt}}(P^{\boldsymbol{\infty}})$.*

As the construction $P \mapsto P^{\boldsymbol{\infty}}$ is clearly invertible, this fact, with Corollary 2.2, gives the following.

Corollary 2.11. *For posets P and P' with maximum and minimum elements, if $\mathcal{D}_{\mathrm{nt}}(P) \cong \mathcal{D}_{\mathrm{nt}}(P')$ then $P \cong P'$.*

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3. Setup, and the definition of $D_{\mathcal{P}}(L)$

For the rest of the paper $\mathcal{P}$ is always a product $\mathcal{P} = \prod \mathcal{C}$ of finite chains $\mathcal{C} = \{C_1, \dots, C_d\}$. When L is a sublattice of $\mathcal{P}$, Theorem 2.9 yields a unique closed family $\mathcal{I}_L$ such that $L = \mathcal{C} \setminus \bigcup \mathcal{I}_L$. The join irreducible elements of $\mathcal{P}$ are clearly of the form

$$\alpha^{(\underline{i})} := (\underbrace{0, 0, \dots, 0, \alpha}_{i}, 0, 0, \dots, 0)$$

for $i \in [d]$ and $\alpha \in C_i$ and the meet-irreducible elements are

$$\beta^{(\overline{j})} := (M_1, M_2, \dots, M_{j-1}, \beta, M_{j+1}, \dots, M_d)$$

for $j \in [d]$ and $\beta \in C_j$, where M_i is the maximum elemement of C_i. The irreducible intervals of $\mathcal{P}$ are exactly the intervals $[\alpha^{(\underline{i})}, \beta^{(\overline{j})}]$ for $i, j \in [d]$, $\alpha \in C_i$, and $\beta \in C_j$. Note that $0^{(\underline{i})} = \mathbf{0}$ for all i, and $\infty^{(\overline{j})} = \boldsymbol{\infty}$ for all j. Given $\mathcal{C}$ and $\mathcal{P} = \prod \mathcal{C}$, let $\mathcal{C}^\infty = J^\infty(\mathcal{P})$ be the subposet of $\mathcal{P}$ induced by $J(\mathcal{P}) \cup \{\mathbf{0}, \boldsymbol{\infty}\}$. As $\boldsymbol{\infty}$ is the element of $\mathcal{P}$ above $M_i^{(\underline{i})}$ and $\mathbf{0}$ is the element below $0^{(\overline{j})}$, it seems reasonable to let $(M_i + 1)^{(\underline{i})} = \boldsymbol{\infty}$ and $(-1)^{(\overline{j})} = \mathbf{0}$. We do this occasionally to make definitions and proofs more compact.

Recall that a poset, and in particular $\mathcal{C}^\infty$, is a *reflexive digraph*: a reflexive relation. We view posets as reflexive digraphs, saying (x, y) is an *arc*, or writing $x \to y$, to mean $x \le y$. A *spanned extension* of $\mathcal{C}^\infty$ is any digraph on the elements of $\mathcal{C}^\infty$, which contains it. A *preorder* is a transitive reflexive digraph.

Definition 3.1 ($D_{\mathcal{P}}(\mathcal{I})$ and $D_{\mathcal{P}}(L)$)**.** For any family $\mathcal{C} = \{C_1, \dots, C_d\}$ of chains and any family of irreducible intervals $\mathcal{I}$ of $\mathcal{P} = \prod \mathcal{C}$, let $D_{\mathcal{P}}(\mathcal{I})$ be the spanned extension of $\mathcal{C}^\infty$ that we get from $\mathcal{C}^\infty$ by letting $(\beta + 1)^{(\underline{j})} \to \alpha^{(\underline{i})}$ for each $[\alpha^{(\underline{i})}, \beta^{(\overline{j})}] \in \mathcal{I}$. For a sublattice L of $\mathcal{P}$, let $D_{\mathcal{P}}(L) = D_{\mathcal{P}}(\mathcal{I}_L)$.

In Proposition 3.6 we will show that because $\mathcal{I}_L$ is closed, $D_{\mathcal{P}}(L)$ is a preorder. The Hasse diagram is not uniquely defined for a preorder, as it is for a poset, but can still be used to depict one– the preorder is still the transitive closure of the Hasse diagram. We therefore depict $D_{\mathcal{P}}(L)$ with a Hasse diagram.

The top half of Figure 2 shows the same lattices $\mathcal{P} = Z_5 \times Z_6$ and $L_1 = \mathcal{P} \setminus [3^{(\underline{2})}, 2^{(\overline{1})}]$ as are shown in Figure 1, but now, instead of $J(\mathcal{P})$ and $J(L_1)$, it shows with them the digraph $\mathcal{C}^\infty$ and its spanned extension $D_{\mathcal{P}}(L_1)$.

An (x, y)*-path* $v_1 \to v_2 \to \cdots \to v_\ell$ in a digraph D is a sequence of elements $x = v_1, v_2, \dots, v_\ell = y$ such that $v_i \to v_{i+1}$ for each i. The following is a straightforward extension of the poset definition of downset to digraphs.

Definition 3.2. A subset $S \subseteq D$ of vertices of a digraph D is a *downset* (or an *initial set*) if for all y in S and all (x, y)-paths in D, x is also in S. A downset set S of D is a *non-trivial downset* if it is a non-empty proper subset of D. For a vertex x of D let $\operatorname{id} x$ be the smallest downset of D containing x.

Definition 3.3 ($\mathcal{D}(D)$ and $\mathcal{D}_{\mathrm{nt}}(D)$)**.** For any digraph D, let $\mathcal{D}(D)$ be the family of downsets of D ordered by inclusion, and $\mathcal{D}_{\mathrm{nt}}(D)$ be the subfamily of non-trivial downsets.

The following simple case of our main theorem is clear from the top example in Figure 2, and indeed, as $\mathcal{P}$ is a lattice and $\mathcal{C}^\infty = J^\infty(L)$, it is essentially just a simple case of Theorem 2.1, using Fact 2.10.

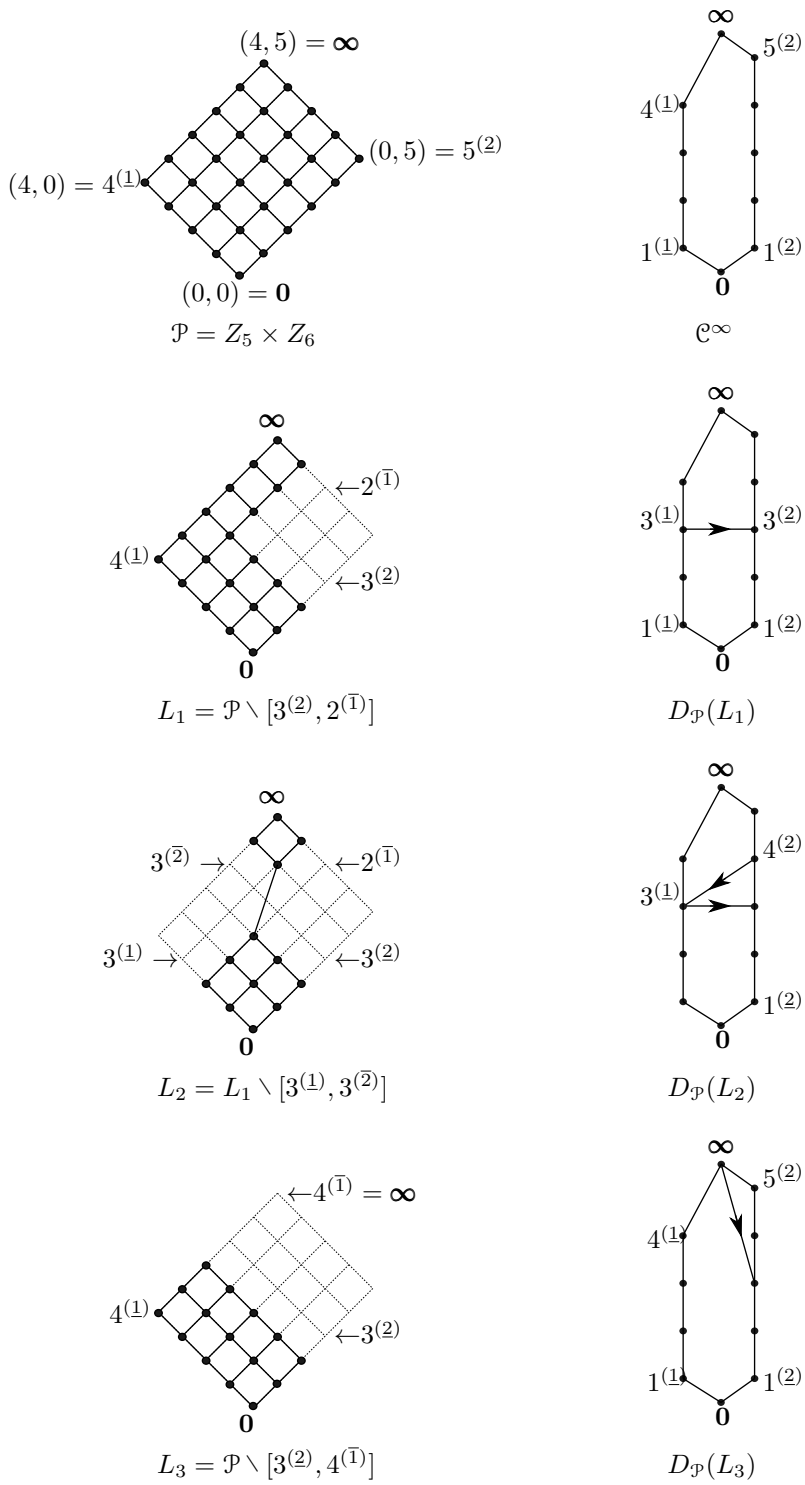

FIGURE 2. Sublattices of $\mathcal{P} = Z_5 \times Z_6$ and their digraphs $D_{\mathcal{P}}$

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Lemma 3.4. *Where $\mathcal{P}$ is a product of chains, the map*

$$\mathbf{T} : \mathcal{P} \to \mathcal{D}_{\mathrm{nt}}(\mathcal{C}^\infty) : x \mapsto \mathrm{id}\, x \cap \mathcal{C}^\infty$$

is an isomorphism. The inverse is $S \mapsto \bigvee_{\mathcal{P}} S$.

Before we prove our main theorem, that $\mathbf{T}$ is an isomorphism from L to $\mathcal{D}_{\mathrm{nt}}(D_{\mathcal{P}}(L))$ for **any** sublattice of $\mathcal{P}$, we motivate the definition of $D_{\mathcal{P}}(L)$ with a couple of examples. It arises from observing what we must do to $\mathcal{C}^\infty$ so that $\mathbf{T}$ remains an isomorphism when we remove some irreducible interval $[\alpha^{(\underline{i})}, \beta^{(\overline{j})}]$ from $\mathcal{P}$.

Referring to the second example in Figure 2, when we remove $[3^{(\underline{2})}, 2^{(\overline{1})}]$ from $\mathcal{P}$, we see that we should add the $3^{(\underline{2})} \geq 3^{(\underline{1})}$ to $\mathcal{C}^\infty$ to maintain the isomorphism. Indeed, this ensures that, for example, the set $\mathbf{T}((2,3)) = \{2^{(\underline{1})}, 1^{(\underline{1})}, \mathbf{0}, 1^{(\underline{2})}, 2^{(\underline{2})}, 3^{(\underline{2})}\}$ is no longer a downset. In general it 'kills' any downset S containing $3^{(\underline{2})}$ but not $3^{(\underline{1})}$. These are exactly the downsets $\mathbf{T}(x)$ for $x \in [3^{(\underline{2})}, 2^{(\overline{1})}]$.

The third example in the figure shows a non-tight sublattice L_2 of $\mathcal{P}$, observe that the corresponding extension $D_{\mathcal{P}}(L_2)$ of $\mathcal{C}^\infty$ is not a poset. The fourth example exhibits the necessity of $\mathbf{0}$ and $\boldsymbol{\infty}$ in $D_{\mathcal{P}}$.

3.1. **The closure of $\mathcal{I}$ and transitivity of $D_{\mathcal{P}}(\mathcal{I})$.** Observe that irreducible intervals of $\mathcal{P} = \prod \mathcal{C}$ of the form $[\alpha^{(\underline{i})}, \beta^{(\overline{i})}]$, where $\beta < \alpha$, are empty. By definition they are contained in any closed family $\mathcal{I}$ of irreducible intervals. This corresponds to $D_{\mathcal{P}}(\mathcal{I})$ containing $\mathcal{C}^\infty$.

Lemma 3.5. *Let $D = D_{\mathcal{P}}(\mathcal{I})$ for some sublattice $L = \mathcal{P} \setminus \bigcup \mathcal{I}$ of $\mathcal{P} = \prod \mathcal{C}$. The following are equivalent for $\alpha \in C_i \in \mathcal{C}$ and $\beta \in C_j \in \mathcal{C}$.*

(i) $[\alpha^{(\underline{i})}, \beta^{(\overline{j})}] \subseteq \bigcup \mathcal{I}$.
(ii) For all $x \in L$, $x_i \geq \alpha \Rightarrow x_j > \beta$.
(iii) There is a $((\beta+1)^{(\underline{j})}, \alpha^{(\underline{i})})$-path in D.

Proof. The equivalence of (i) and (ii) is immediate as both are equivalent to the statement that there are no elements $x \in L$ with $\alpha \leq x_i$ and $x_j \leq \beta$. So we show the equivalence of (ii) and (iii).

On the one hand, assume that

$$(\beta+1)^{(\underline{s_\ell})} = \alpha_{s_\ell}^{(\underline{s_\ell})} \to \cdots \to \alpha_{s_2}^{(\underline{s_2})} \to \alpha_{s_1}^{(\underline{s_1})}$$

is an $((\beta+1)^{(\underline{s_\ell})}, \alpha_{s_1}^{(\underline{s_1})})$-path in D. By definition of $D = D_{\mathcal{P}}$ this means that $[\alpha_{s_i}{}^{(\underline{s_i})}, (\alpha_{s_{i+1}} - 1)^{(\overline{s_{i+1}})}] \in \mathcal{I}$ for each i. Let $x \in L$ be such that $x_{s_1} \geq \alpha_{s_1}$. By the equivalence of (i) and (ii) this implies $x_{s_2} > \alpha_{s_2} - 1$, which in turn implies implies $x_{s_3} > \alpha_{s_3} - 1$, etc., until we get that $x_{s_\ell} > \alpha_{s_\ell} - 1 = \beta$, as needed.

On the other hand, assume that there is no such path from $\alpha_{s_\ell}^{(\underline{s_\ell})}$ to $\alpha_{s_1}^{(\underline{s_1})}$. We will find x in L with $x_{s_1} \geq \alpha_{s_1}$ and $x_{s_\ell} < \alpha_{s_\ell} = \beta + 1$. Indeed, for $k \in [d]$ let x_k be the maximum value for which there is a path in D from $x_k^{(\underline{k})}$ to $\alpha_{s_1}^{(\underline{s_1})}$. If no such path exists, let x_k be 0. Clearly $x_{s_1} \geq \alpha_{s_1}$, and by the assumption that there is no path from $\alpha_{s_\ell}^{(\underline{s_\ell})}$ to $\alpha_{s_1}^{(\underline{s_1})}$ we have that $x_{s_\ell} < \alpha_{s_\ell}$. We have just to show that x is in L.

If x is not in L, then it is in some $[\alpha_i^{(\underline{i})}, \beta_j^{(\overline{j})}] \in \mathcal{I}$. As $[\alpha_i^{(\underline{i})}, \beta_j^{(\overline{j})}] \in \mathcal{I}$, we have that $(\beta_j + 1)^{(\underline{j})} \to \alpha_i^{(\underline{i})}$ is in D. As $x \in [\alpha_i^{(\underline{i})}, \beta_j^{(\overline{j})}]$, we have that $\alpha_i \leq x_i$ and $x_j \leq \beta_j$. Now by the choice of x, $\alpha_i \leq x_i$ implies that there is an path from $\alpha_i^{(\underline{i})}$ to $\alpha_{s_i}^{(\underline{s_i})}$,

so the arc $((\beta_j+1)^{(\underline{j})}, \alpha_{s_i}^{(\underline{s_i})})$ gives us a path from $(\beta_j+1)_{s_j}^{(\underline{s_j})}$ to $\alpha_{s_i}^{(\underline{s_i})}$. But then $\beta_j+1 \le x_j$ which contradicts $x_j \le \beta_j$. □ □

As $\mathcal{I}_L$ is closed, the following proposition allows us to assume that the digraph $D_{\mathcal{P}}(L)$, for any sublattice L of $\mathcal{P}$, is a preorder.

Proposition 3.6. *Let $D = D_{\mathcal{P}}(\mathcal{I})$ for some sublattice $\mathcal{P}\setminus\bigcup\mathcal{I}$ of $\mathcal{P}$. Then $\mathcal{I}$ is closed if and only if D is transitive.*

Proof. On the one hand, assume that $\mathcal{I}$ is closed. Then we can replace (i) of Lemma 3.5 with $[\alpha^{(\underline{i})}, \beta^{(\overline{j})}] \in \mathcal{I}$. Now that $\gamma^{(\underline{k})} \to \beta^{(\underline{j})}$ and $\beta^{(\underline{j})} \to \alpha^{(\underline{i})}$ in D gives, by Lemma 3.5, that $[\alpha^{(\underline{i})}, \gamma^{(\overline{k})}] \in \mathcal{I}$ which means $\gamma^{(\underline{k})} \to \alpha^{(\underline{i})}$ in D. So D is transitive.

On the other hand, assume that $\mathcal{I}$ is not closed. Then there is some $[\alpha^{(\underline{i})}, \beta^{(\overline{j})}]$ in $\bigcup\mathcal{I}$ but not in $\mathcal{I}$. So there is a $(\alpha^{(\underline{i})}, \beta^{(\overline{j})})$-path in D, while $(\alpha^{(\underline{i})}, \beta^{(\overline{j})})$ is not in D. □ □

4. Main Theorem: Sublattices of $\mathcal{P}$ as downset lattices

As with posets, it is clear for digraphs that unions and intersections of downsets are downsets, so $\mathcal{D}_{\mathrm{nt}}(D)$ is indeed a poset for any digraph D. We only need this when D is a spanned extension of some $\mathcal{C}^\infty$, but in this case, we get more.

Theorem 4.1. *Let L be a sublattice of a product of chains $\mathcal{P}$, and $D = D_{\mathcal{P}}(L)$. The map $\mathbf{T} : L \to \mathcal{D}_{\mathrm{nt}}(D) : x \mapsto \mathrm{id}\, x \cap \mathcal{C}^\infty$ is a lattice isomorphism.*

Proof. As it simplifies induction, we prove the slightly more general result that $\mathbf{T}$ is an isomorphism for $D = D_{\mathcal{P}}(\mathcal{I})$ where $\mathcal{I}$ is any family of irreducible intervals of $\mathcal{P}$ such that $L = \mathcal{P}\setminus\bigcup\mathcal{I}$. In the case that $\mathcal{I}$ is empty, or contains only empty families, we have $L = \mathcal{P}$ and $D = \mathcal{C}^\infty$, so the isomorphism is given in Lemma 3.4.

In the general case, we first observe that $\mathcal{D}_{\mathrm{nt}}(D)$ is a subposet of $\mathcal{P} = \mathcal{D}_{\mathrm{nt}}(\mathcal{C}^\infty)$. To see that it is a subset, observe that adding arcs to D can only remove downsets, not create new ones: if a downset in D contains $x^{(\underline{i})}$, then it must contain $\alpha^{(\underline{i})}$ for all $\alpha < x$. So any downset set of D is the union, over its elements x, of the downsets $\mathrm{id}\, x$ they generate in $\mathcal{C}^\infty$. As $\mathcal{D}_{\mathrm{nt}}(D)$ is also ordered by inclusion, its order is induced from $\mathcal{D}_{\mathrm{nt}}(\mathcal{C}^\infty)$, so is a subposet.

Now it is enough to show that $\mathbf{T} : L \to \mathcal{D}_{\mathrm{nt}}(D)$ a bijection. We do this by induction on the size of $\mathcal{I}$. Let $[\alpha^{(\underline{i})}, \beta^{(\overline{j})}] \in \mathcal{I}$, $\mathcal{I}' = \mathcal{I}\setminus[\alpha^{(\underline{i})}, \beta^{(\overline{j})}]$, $L' = \mathcal{P}\setminus\bigcup\mathcal{I}'$, and $D' = D_{\mathcal{P}}(\mathcal{I}')$. By induction we have that $\mathbf{T} : L' \to \mathcal{D}_{\mathrm{nt}}(D')$ is a bijection.

We must show for $\mathbf{T}(x) \in \mathcal{D}_{\mathrm{nt}}(D')$, that $\mathbf{T}(x) \notin \mathcal{D}_{\mathrm{nt}}(D)$ if and only if $x \in [\alpha^{(\underline{i})}, \beta^{(\overline{j})}]$. Now $\mathbf{T}(x)$ being a downset in D', but not in D which we get from D' by adding the arc $((\beta+1)^{(\underline{j})}, \alpha^{(\underline{i})})$, means exactly that $\alpha^{(\underline{i})} \in \mathbf{T}(x)$ and $(\beta+1)^{(\underline{j})} \notin \mathbf{T}(x)$. This is to say $\alpha \le x_i$ but $(\beta+1) > x_j$, which means that $x \in [\alpha^{(\underline{i})}, \beta^{(\overline{j})}]$, as needed. □ □

By the theorem, we have that any sublattice of $\mathcal{P}$ can be expressed as the lattice $\mathcal{D}_{\mathrm{nt}}(D)$ of non-trivial downsets of some spanned extension D of $\mathcal{C}^\infty$. On the other hand, it is clear that for a spanned extension D of $\mathcal{C}^\infty$ the family

$$\mathcal{I}_D = \{[\alpha^{(\underline{i})}, \beta^{(\overline{j})}] \mid ((\beta+1)^{(\underline{j})}, \alpha^{(\underline{i})}) \in D\setminus\mathcal{C}^\infty\}$$

is such that $\mathcal{D}_{\mathrm{nt}}(D)$ is the sublattice $\mathcal{P}\setminus\bigcup\mathcal{I}_D$ of $\mathcal{P}$. While several digraphs D may yield the same sublattice of $\mathcal{P}$, as may families of intervals have the same union, it

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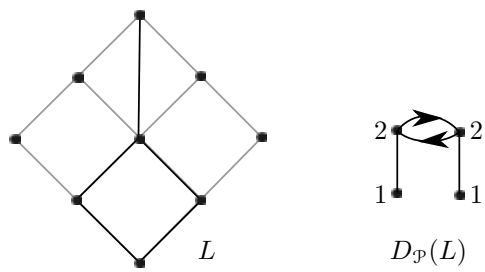

FIGURE 3. Non-tight subdirect sublattice of $Z_3 \times Z_3$

is clear that there is a unique closed family of intervals defining a given sublattice. So by Proposition 3.6 we get the following.

Corollary 4.2. *The map $L \mapsto D_{\mathcal{P}}(L)$ gives a one-to-one correspondence between the sublattices of $\mathcal{P}$ and the spanned preorder extensions of $\mathcal{C}^{\infty}$.*

5. Classification of sublattices

Recall that a sublattice L of $\mathcal{P}$ is a $\{\mathbf{0}, \boldsymbol{\infty}\}$-sublattice if it contains the extremal elements $\mathbf{0}_{\mathcal{P}}$ and $\boldsymbol{\infty}_{\mathcal{P}}$ of $\mathcal{P}$. It is *subdirect* if for each $i \in [d]$, the projection $\pi_i : L \to C_i$ is surjective; this is necessarily a $\{\mathbf{0}, \boldsymbol{\infty}\}$-sublattice. A $\{\mathbf{0}, \boldsymbol{\infty}\}$-sublattice is *tight* if its covers are covers of $\mathcal{P}$.

It was shown in [6] that every tight sublattice of a product of chains is subdirect. The converse was also claimed, but the proof was flawed: indeed, the lattice $L = Z_3 \times Z_3 \setminus ([2^{(\underline{1})}, 1^{(\overline{2})}] \cup [2^{(\underline{2})}, 1^{(\overline{1})}])$ shown in Figure 3 is a subdirect sublattice of $Z_3 \times Z_3$, but not tight. One notices in this example that $D_{\mathcal{P}}(L)$ has a cycle. We will see that this is indicative of non-tight sublattices.

The *height* of a lattice is the length of a maximum ascending chain of covers from $\mathbf{0}$ to $\boldsymbol{\infty}$. It is well known that for a distributive lattice, every cover is in a maximum ascending chain. It follows that a sublattice L of $\mathcal{P}$ is tight if and only if L and $\mathcal{P}$ have the same height. An arc (x, y) in a digraph D is an *in-arc* of y and an *out-arc* of X.

Lemma 5.1. *Let L be a sublattice of $\mathcal{P}$ and $D = D_{\mathcal{P}}(L)$.*

- *(i) L is a $\{\mathbf{0}, \boldsymbol{\infty}\}$-sublattice if and only if $\mathbf{0}$ is a source (has no in-arcs) in D and $\boldsymbol{\infty}$ is a sink (has no out-arcs).*
- *(ii) L is subdirect if and only if D has no* down edges*: those of the form $\alpha^{(\underline{i})} \to (\alpha - 1)^{(\underline{i})}$.*
- *(iii) L is tight if and only if D is a poset.*

Proof. We have by Theorem 4.1 that $\mathbf{T} : L \to \mathcal{D}_{\mathrm{nt}}(D)$ is an isomorphism.
(i) Clearly $\mathbf{T}(\mathbf{0}) = \{\mathbf{0}\}$ is a downset of D if and only if $\mathbf{0}$ is a source in D, and $\mathbf{T}(\boldsymbol{\infty}) = \mathcal{C}^{\infty} \setminus \{\boldsymbol{\infty}\}$ is a downset if and only if $\boldsymbol{\infty}$ is a sink. The result follows.

(ii). We have $\alpha^{(\underline{i})} \to (\alpha-1)^{(\underline{i})}$ in D if and only if there is no downset $\mathbf{T}(x)$ of D containing $(\alpha-1)^{(\underline{i})}$ but not $\alpha^{(\underline{i})}$. This is if and only if there is no element $x \in L$ with $x_i = \alpha - 1$.
(iii). First assume that L is tight, and assume, towards contradiction that D contains a cycle $\alpha_1^{(\underline{i_1})} \to \alpha_2^{(\underline{i_2})} \cdots \to \alpha_\ell^{(\underline{i_\ell})} \to \alpha_1^{(\underline{i_1})}$, for some $\ell \geq 2$. We show that no vertex x in L has $x_{i_1} = \alpha_1$, which contradicts the fact that L is tight. Indeed, if x did have $x_{i_1} = \alpha_1$, then $x_{i_1} \geq \alpha_1$ so by Lemma 3.5 $x_{i_2} > \alpha_2$ so $x_{i_3} > \alpha_3$ etc., and we get that $x_{i_1} > \alpha_1$, a contradiction.

On the other hand, assume that D has no cycles. Then its vertices can be ordered so that all arcs go up. Visiting vertices from the bottom of this ordering, one at a time, we get an ascending walk in $L \cong \mathcal{D}_{\mathrm{nt}}(D)$ from $\mathbf{0}$ to $\boldsymbol{\infty}$ of size $|D|$, showing that L has height equal to the height of $\mathcal{P}$. Thus L is a tight sublattice. □ □

For any preorder D with named elements $\mathbf{0}$ and $\boldsymbol{\infty}$, let D^* be the subgraph induced by $D \setminus \{\mathbf{0}, \boldsymbol{\infty}\}$. A spanned extension $D_{\mathcal{P}}(L)$ of $\mathcal{C}^\infty$ yields a spanned extension $D^*_{\mathcal{P}}(L)$ of $\mathcal{C}^* := (\mathcal{C}^\infty)^*$, so with Lemma 5.1(i), Corollary 4.2 restricts to the following. We use Lemma 5.1(iii) in the second statement.

Corollary 5.2. *The map $L \mapsto D^*_{\mathcal{P}}(L)$ gives a one-to-one correspondence between the $\{\mathbf{0}, \boldsymbol{\infty}\}$-sublattices of $\mathcal{P}$ and the spanned preorder extensions of $\mathcal{C}^*$. It gives a one-to-one correspondence between the tight sublattices of $\mathcal{P}$ and the spanned poset extensions of $\mathcal{C}^*$.*

Lemma 5.1(i) also gives us the following.

Lemma 5.3. *Let L be a $\{\mathbf{0}, \boldsymbol{\infty}\}$-sublattice of $\mathcal{P}$ and $D = D_{\mathcal{P}}(L)$. Then the map $\mathcal{D}_{\mathrm{nt}}(D) \to \mathcal{D}(D^*) : S \mapsto S \setminus \{\mathbf{0}\}$ is an isomorphism.*

Proof. We show that $S \mapsto S \setminus \{\mathbf{0}\}$ is a bijection; as the orders in $\mathcal{D}_{\mathrm{nt}}(D)$ and $\mathcal{D}(D^*)$ are subset orders, it then clearly preserves order.

As $\{\mathbf{0}\}$ is in any non-trivial downset D and has no in-arcs, it is clear that for any non-trivial downset S of D, $S \setminus \{\mathbf{0}\}$ is downset of D^*. We must show that there are no other downsets of D^*. Assume S is a downset in D^* but $S \cup \{\mathbf{0}\}$ is not a downset of D. Then S must have some in-arc in D that does not exist in D^*. This can only be an arc from $\boldsymbol{\infty}$ but by Lemma 5.1(i), there are no such arcs. □ □

Using Lemma 5.3 to extend the isomorphism $\mathbf{T} : L \to \mathcal{D}_{\mathrm{nt}}(D)$ of Theorem 4.1 to an isomorphism $\mathbf{S} : L \to \mathcal{D}_{\mathrm{nt}}(D) \to \mathcal{D}(D^*)$, we get the following.

Corollary 5.4. *Let L be a $\{\mathbf{0}, \boldsymbol{\infty}\}$-sublattice of $\mathcal{P} = \prod \mathcal{C}$ and let $D^* = D^*_{\mathcal{P}}(L)$. The map $\mathbf{S} : L \to \mathcal{D}(D^*) : x \mapsto \mathrm{id}\, x \cap \mathcal{C}$ is a lattice isomorphism.*

That Corollary 5.4 and so Theorem 4.1 are extensions of Theorem 2.1 become more explicit with the following observation.

Lemma 5.5. *If L is a tight sublattice of $\mathcal{P}$, then $D^*_{\mathcal{P}}(L) \cong J(L)$.*

Proof. If L is a tight sublattice of $\mathcal{P}$ then $D_{\mathcal{P}}(L)$ is a poset by Lemma 5.1(iii), and so $D^*_{\mathcal{P}}(L)$ is too. Thus $L \cong \mathcal{D}_{\mathrm{nt}}(D_{\mathcal{P}}(L)) \cong \mathcal{D}(D^*_{\mathcal{P}}(L))$ using Lemma 5.3, and the statement follows by Corollary 2.2. □ □

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6. Back to Embeddings

We now view a sublattice L as the image of an embedding $e : L \to \mathcal{P}$ and the subgraph $\mathcal{C}^\infty$ of D as the image of a bijective homomorphism $\phi : \mathcal{C}^\infty \to D$. As such, Corollary 4.2 gives a correspondence between embeddings of lattices into $\mathcal{P}$ and surjective homomorphisms of $\mathcal{C}^\infty$ to preorders D.

Formally, a *pointed union of chains* is any poset that we can form from a parallel union of disjoint chains by identifying the minimum elements of the chains and adding a new maximum element $\boldsymbol{\infty}$– that is, any poset that can be represented as $\mathcal{C}^\infty$ for some product $\mathcal{P} = \prod \mathcal{C}$ of chains. A *pointed chain decomposition* of a preorder D is a bijective homomorphism $\phi : \mathcal{C}^\infty \to D$ from a pointed union of chains.

Given a pointed chain decomposition $\phi : \mathcal{C}^\infty \to D$ of D. the map $\mathbf{T}$ of Theorem 4.1 becomes $\mathbf{T} : L \to \mathcal{D}_{\mathrm{nt}}(D) : x \mapsto \phi(\mathrm{id}\, x \cap \mathcal{C}^\infty)$ and its inverse becomes $\mathbf{T}^{-1} : S \mapsto \bigvee_{\mathcal{P}} \phi^{-1}(S)$. This gives the following.

Corollary 6.1. *For any pointed chain decomposition $\phi : \mathcal{C}^\infty \to D$ of a preorder D the map $e_\phi : S \mapsto \bigvee_{\mathcal{P}} \phi^{-1}(S)$ is an embedding of $\mathcal{D}_{\mathrm{nt}}(D)$ into $\mathcal{P}$.*

By Lemma 5.1, e_ϕ is $\{\mathbf{0}, \boldsymbol{\infty}\}$-embedding if and only if $\mathbf{0}$ is a source of D and $\boldsymbol{\infty}$ is a sink. In this case $\phi^{-1}(\mathbf{0}) = \{\mathbf{0}\}$ and $\phi^{-1}(\boldsymbol{\infty}) = \{\boldsymbol{\infty}\}$. Similar to the observation we made before Corollary 5.2, this means that $\phi(\mathcal{C}^*)$ is a chain decomposition of D^*. As $\mathcal{C}^* = (\mathcal{C}^\infty)^*$ is the union of the chains of $\mathcal{C}$, with their zero elements removed, and $\mathcal{C}_0^*$ is the family of chains we get from the chains in $\mathcal{C}^*$ by adding a new minimum element to each, $\mathcal{C}_0^* = \mathcal{C}$.

The following is thus a corollary of Corollary 5.4. Identifying $\phi(\mathcal{C}^*)$ with $\mathcal{C}^*$, it is a generalization of Theorem 2.3.

Corollary 6.2. *For any chain decomposition $\phi : \mathcal{C}^* \to D^*$ of a preorder D^* let $\mathcal{C} = \mathcal{C}_0^*$, the map $e_\phi : S \mapsto \bigvee_{\mathcal{P}} S$ is a $\{\mathbf{0}, \boldsymbol{\infty}\}$-embedding of $\mathcal{D}(D^*)$ into $\mathcal{P} = \prod \mathcal{C}$.*

Now, for given distributive lattice L, there are several preorders D for which $L \cong \mathcal{D}_{\mathrm{nt}}(D)$, so which yield embeddings of L into a product of chains. To get back to a canonical D, we first must take a quotient.

6.1. A Canonical Quotient of $D_{\mathcal{P}}(L)$.

The *non-preference* relation '$\sim$' defined on a preorder D by letting $x \sim y$ if x and y are in a directed cycle is an equivalence relation. The quotient $P_D := D/\sim$ is well known to be a poset. Classes of P_D satisfy $[a] \leq [b]$ if and only if $a \leq b$ in D.

Now the *quotient homomorphism* $q : D \to P_D : a \mapsto [a]$ induces maps between 2^D and 2^{P_D}. For $T \subseteq D$, we let $[T] := \{[x] \mid x \in T\} \subseteq P_D$, and for $S \subseteq P_D$ we let $\bigcup S = \bigcup_{[x] \in S} [x]$. Clearly $[\bigcup S] = S$ and $\bigcup [T] = T$.

Lemma 6.3. *Let D be a preorder, then $\mathcal{D}_{\mathrm{nt}}(D) \to \mathcal{D}_{\mathrm{nt}}(P_D) : T \mapsto [T]$ is an isomorphism with inverse $S \mapsto \bigcup S$.*

Proof. Clearly $[T] \in P_D$ is a downset if $T \subset D$ is, and $\bigcup S \subset D$ is if P_D is, so $T \mapsto [T]$ is a bijection. That is preserves order is also clear as the order on both posets is inclusion, and $[T]$ simply partitions a set T with respect to an underlying fixed partition of D. □ □

Recall that for a lattice L, $J^\infty(L)$ is the subgraph induced by $J(L) \cup \{\mathbf{0}, \boldsymbol{\infty}\}$. By Fact 2.10 we have $\mathcal{D}_{\mathrm{nt}}(J^\infty(L)) \cong \mathcal{D}(J(L)) \cong L$, and by Theorem 4.1 and Lemma

6.3 that $L \cong \mathcal{D}_{\mathrm{nt}}(D)) \cong \mathcal{D}_{\mathrm{nt}}(P_D)$ where $D = D_{\mathcal{P}}(L)$. So by Corollary 2.11 we get the following.

Corollary 6.4. *For a sublattice L of a product $\mathcal{P} = \prod \mathcal{C}$, $P_D \cong J^\infty(L)$.*

6.2. **Embeddings of L and homomophisms to $J^\infty(L)$.** Let $e : L \to \mathcal{P}$ be an embedding of a lattice into a product of chains, so Corollary 4.2 yields a corresponding spanned extension $D = D_{\mathcal{P}}(e(L))$ of $\mathcal{C}^\infty$, which we again view as a pointed chain decompositon $\phi_0 : \mathcal{C}^\infty \to D$. Composing with $q : D \to P_D \cong J^\infty(L) =: J^\infty$, we get a surjective homomorphism $\phi_e = q \circ \phi_0 : \mathcal{C}^\infty \to J^\infty$.

Using the isomorphsim $L \cong \mathcal{D}_{\mathrm{nt}}(J^\infty)$ we get an explicit formula for ϕ_e, and show that $e \mapsto \phi_e$ is a one-to-one correspondence. For this section we denote a principle downset $\mathrm{id}\, x$ of J^∞ by $\langle x]$, and its unique predecessor $\mathrm{id}\, x \setminus \{x\}$ in $\mathcal{D}_{\mathrm{nt}}(J^\infty)$ by $\langle x)$. Similarily we use $\langle \cdot]_{\mathcal{P}}$ and $\langle \cdot)_{\mathcal{P}}$ for downsets in $\mathcal{P}$.

For any surjective homomorphism $\phi : \mathcal{C}^\infty \to J^\infty$, let

$$e_\phi : \mathcal{D}_{\mathrm{nt}}(J^\infty) \to \mathcal{P} : \langle x] \mapsto \bigvee \phi^{-1}(\langle x]). \tag{1}$$

For an embedding $e : \mathcal{D}_{\mathrm{nt}}(J^\infty) \to \mathcal{P}$, let $\phi_e : \mathcal{C}^\infty \to J^\infty$ be defined by

$$\phi_e^{-1}(x) = [\langle e\,\langle x]]_{\mathcal{P}} \setminus \langle e\,\langle x])_{\mathcal{P}}] \cap \mathcal{C}^\infty. \tag{2}$$

We are ready for our generalisation of Corollary 2.6.

Theorem 6.5. *For any distributive lattice L, let $J^\infty = J^\infty(L)$. The maps $\phi \to e_\phi$ and $e \to \phi_e$ defined in equations* (1) *and* (2) *give a one-to-one correspondence between embeddings of $\mathcal{D}_{\mathrm{nt}}(J^\infty) \cong L$ into products of chains, and surjective homomorphisms of pointed union of chains to J^∞.*

Proof. For a downset $\langle x]$ we have $\phi_e^{-1}\langle x] = \langle e\,\langle x]]_{\mathcal{P}} \cap \mathcal{C}^\infty$, and so $e = e_{\phi_e}$ is immediate. Indeed, recalling for the last equality that $p \to \mathrm{id}\, p \cap \mathcal{C}^\infty$ and $S \to \bigvee S$ are inverse :

$$e_{\phi_e}\langle x] = \bigvee \phi_e^{-1}\langle x] = \bigvee \langle e\,\langle x]]_{\mathcal{P}} \cap \mathcal{C}^\infty = e\,\langle x].$$

To see that $\phi_{e_\phi} = \phi$, let x be in J^∞. Using in the third line that $\langle p] \setminus \langle p) = p$ for any p, we get

$$\begin{aligned}
\phi_{e_\phi}^{-1}(x) &= \left[\langle e_\phi\,\langle x]]_{\mathcal{P}} \setminus \langle e_\phi\,\langle x])_{\mathcal{P}}\right] \cap \mathcal{C}^\infty \\
&= \left[\left\langle \bigvee \phi^{-1}\langle x]\right]_{\mathcal{P}} \setminus \left\langle \bigvee \phi^{-1}\langle x]\right)_{\mathcal{P}}\right] \cap \mathcal{C}^\infty \\
&= \left[\bigvee \phi^{-1}\langle x]\right] \cap \mathcal{C}^\infty \\
&= \phi^{-1}(x).
\end{aligned}$$

□ □

6.3. **Classification of Embeddings.** For an embedding $e : L \to \mathcal{P}$, we have a surjective homomorphism $\phi_e = q \circ \phi_0 : \mathcal{C}^\infty \to D_{\mathcal{P}}(e(L)) \to J^\infty$. We observed for Corollary 6.2 that when e is a $\{\mathbf{0}, \boldsymbol{\infty}\}$-embedding then be removing extremal elements, ϕ_0 induces a homorphism of $\mathcal{C}^*$ to D^*. This extends by q to a surjection $\phi_e : \mathcal{C}^* \to J(e(L))$. Again by Lemma 5.1, e is subdirect if and only if $D = D_{\mathcal{P}}(e(L))$ has no down edges, which is true if and only if all ϕ_e is injective on all chains of $\mathcal{C}^*$. It is tight if and only if D contains no cycles, which is true if and only if ϕ_e is injective.

We have thus showed the following corollary to Theorem 6.5.

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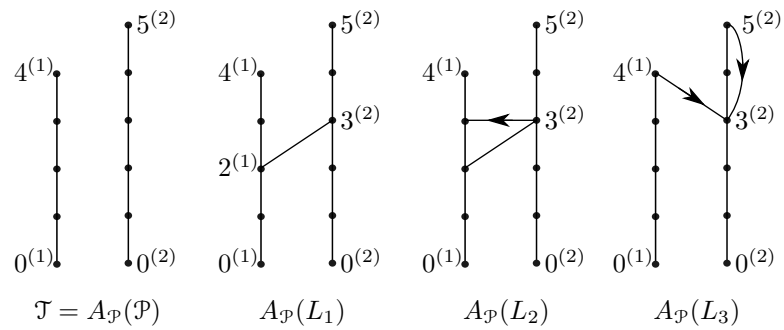

FIGURE 4. Sublattices of $\mathcal{P} = Z_4 \times Z_5$ and their digraphs $A_{\mathcal{P}}$

Corollary 6.6. *The constructions $e \to \phi_e$ and $\phi \to e_\phi$ give a one-to-one correspondence between full embeddings of L into products of chains, and surjective homomorphism of disjoint unions of chains to $J(L)$. Subdirect embeddings correspond with homomorphisms that are injective on each chain, and tight embeddings correspond with bijections– so-called chain decompositions.*

7. Sublattices of $\mathcal{P}$ as Downset or Independent set Lattices

In this section, when we have a product $\mathcal{P} = \prod C_i$ of a family $\mathcal{C} = \{C_1, \ldots, C_d\}$ of chains, $\mathcal{T}$ will be the *union of tournaments* we get from the parallel sum of the chains in $\mathcal{C}$ by removing the loops from all vertices. As we will be now referring to elements of a chain C_i individually, we afix a superscript, for example, denoting the element 3 or x_j, of C_i by $3^{(i)}$ or $x_j^{(i)}$ respectively.

Definition 7.1 ($A_{\mathcal{P}}(\mathcal{I})$ and $A_{\mathcal{P}}(L)$)**.** For a family $\mathcal{C} = \{C_1, \ldots, C_d\}$ of chains and a family of irreducible intervals $\mathcal{I}$ of $\mathcal{P} = \prod \mathcal{C}$, let $A_{\mathcal{P}}(\mathcal{I})$ be the digraph we get from $\mathcal{T}$ by adding the arc $(\beta^{(j)}, \alpha^{(i)})$ for each $[\alpha^{(\underline{i})}, \beta^{(\overline{j})}] \in \mathcal{I}$. For a sublattice L of $\mathcal{P}$, let $A_{\mathcal{P}}(L) = A_{\mathcal{P}}(\mathcal{I}_L)$. (See Figure 4 for examples of $A_{\mathcal{P}}(L)$ corresponding to the sublattices L of $\mathcal{P}$ in Figure 2. The digraph $A_{\mathcal{P}}(L)$ is still the transitive closure of the depicted digraph, where all unoriented arcs are oriented up, but it is no longer reflexive.)

A directed path $x_1 \to x_2 \to \cdots \to x_d$ in a digraph is *non-trivial* if $d \geq 1$. Observe that the directed path x is trivial, while the directed path, or *loop*, $x \to x$, is non-trivial.

For a vertex $x = (x_1, \ldots, x_d)$ of $\mathcal{P}$ we let I_x be the set

$$I_x := \{x_1^{(1)}, \ldots, x_d^{(d)}\}.$$

Definition 7.2. A subset I of a digraph D is *independent* if there is no non-trivial xy-path in D for x and y in I.

Note that this is a much stronger notion of independence than is usually defined for digraphs, it implies independence in the transitive closure of the graph. In fact, no vertex that is in a cycle, including looped vertices, can be in an independent set. So for a digraph like $\mathcal{T}$ which we got by removing loops from a poset, the independent sets are exactly the antichains of the poset. For $\mathcal{T}$ in particular, the d element independent sets are exactly I_x for $x \in \mathcal{P}$.

Definition 7.3 ($\mathbb{A}_d(A)$)**.** Let $\mathbb{A}_d(A)$ be the family of independent sets of size d of a digraph A, and order it as follows. For $I, I' \in \mathbb{A}_d(A)$, let $I \leq I'$ if for each $a' \in I'$ there is an aa'-path in A with $a \in I$.

When restricting to spanned extensions A of $\mathcal{T}$, for which all d-element subsets are of the form I_x, this ordering has a simpler definition, observed in [5] in the case that A is a poset.

Lemma 7.4. *For a spanned extension A of $\mathcal{T}$ and $I_x, I_y \in \mathbb{A}_d(A)$, we have $I_x \leq I_y$ if and only $x_i \leq y_i$ for all i.*

Proof. The 'if' direction is immediate from Definition 7.3 as all arcs of $\mathcal{T}$ are in A. For the 'only if' direction, assume that $I_x \leq I_y$, and let $i \in [d]$. We must show that $x_i \leq y_i$. Assume that it is not, so $y_i < x_i$, and so there is a directed path in A from y_i to x_i. As $I_x \leq I_y$ there is a path in A from x_i to y_j for some $j \in [d]$. If $i = j$ the x_i and y_i are in a directed cycle, contradicting the fact that x_i is in an independent set. If $i \neq j$ then there is a path from y_i to x_i and one from x_i to y_j, contradicting the fact that y_i and y_j are in an independent set. □ □

The following is thus clear.

Lemma 7.5. *Where $\mathcal{P} = \prod \mathcal{C}$, the map $I : \mathcal{P} \to \mathbb{A}_d(\mathcal{T}) : x \mapsto I_x$ is an isomorphism.*

Now, consider removing an irreducible interval to make a sublattice L of $\mathcal{P}$. Figure 4 shows which arcs we need to add to $\mathcal{T} = A_{\mathcal{P}}(\mathcal{P})$ to make a digraph $A_{\mathcal{P}}(L)$ so that the isomorphism $L \cong \mathbb{A}_d(A_{\mathcal{P}}(L))$ is preserved. Recall that $L_1 = \mathcal{P} \setminus [3^{(\underline{2})}, 2^{(\overline{1})}]$. To ensure that, for example, $I_{(2,3)} = \{2^{(1)}, 3^{(2)}\}$ is no longer independent, we add the arc $(2^{(1)}, 3^{(2)})$ to $\mathcal{T}$. This also ensures that such sets an $I_{(2,4)}$ and $I_{(1,5)}$ are no longer independent.

Remark 7.6. Though L_3 is $\mathcal{P} \setminus [3^{(\underline{2})}, 4^{(\overline{1})}]$ we added the arcs $(4^{(1)}, 3^{(2)})$ and $(5^{(2)}, 3^{(2)})$. This is because the intervals $[3^{(\underline{2})}, 4^{(\overline{1})}]$ and $[3^{(\underline{2})}, 5^{(\overline{2})}]$ are the same.

The proof of the following is very similar to that of Theorem 4.1.

Theorem 7.7. *Let L be sublattice of $\mathcal{P} = \prod \mathcal{C}$, where $\mathcal{C} = \{C_1, \ldots, C_d\}$ is a family of disjoint chains, and let $A = A_{\mathcal{P}}(L)$. The map*

$$I : L \to \mathbb{A}_d(A) : x \mapsto I_x := \{x_1^{(1)}, \ldots, x_d^{(d)}\}$$

is a lattice isomorphism.

Proof. As adding arcs to $\mathcal{T}$ or a spanned extension cannot not create new independent sets, we have that $\mathbb{A}_d(A)$ is a subset of $\mathcal{P} = \mathbb{A}_d(\mathcal{T})$; and so by Lemma 7.4 it is a subposet.

Thus it is enough to show that $I : L \to \mathbb{A}_d(A)$ is a bijection. We do this by induction on the size of A. In the case that $A = \mathcal{T}$, we have $L = \mathcal{P}$, so the isomorphism is given in Lemma 7.5. Now let $A = A' \cup \{(\beta^{(j)}, \alpha^{(i)})\}$.

We must show that an independent set $I_x \in \mathbb{A}_d(A')$ is not in $\mathbb{A}_d(A)$ if and only if x is in $[\alpha^{(\underline{i})}, \beta^{(\overline{j})}]$. That is, we must show that $I_x \in \mathbb{A}_d(A') \setminus \mathbb{A}_d(A)$ if and only if $x_i \geq \alpha$ and $\beta \geq x_j$. For $I_x \in \mathbb{A}_d(A')$, clearly $I_x \notin \mathbb{A}_d(A)$ if and only if $(\beta^{(j)}, \alpha^{(i)})$ completes a path between x_j and x_i, so if and only if $x_i \geq \alpha$ and $\beta \geq x_j$, as needed. □ □

Taking the inverse isomorphism I^{-1} in Theorem 7.7 we get an 'independent set analogue' to Corollary 6.1. A special case of it was implicit in [5].

Corollary 7.8. *For any spanned extension A of $\mathcal{T}$ the map $I^{-1} : \mathbb{A}_d(A) \to \mathcal{P}$ is an embedding of $\mathbb{A}_d(A)$ into $\mathcal{P}$.*

7.1. **Consequences.** It is not hard to see that for a given family $\mathcal{I}$ of irreducible intervals, we get from $D_{\mathcal{P}}(\mathcal{I})$ to $A_{\mathcal{P}}(\mathcal{I})$ by 'dropping the source vertex of every arc by one'. We do this for closed families $\mathcal{I}$ with the following construction.

Definition 7.9. For a spanned extension D of $\mathcal{C}^\infty$, let $A = K_{\mathcal{P}}(D)$ be the digraph on the vertices of $\mathcal{T}$ with arcset

$$\{(\beta^{(j)}, \alpha^{(\underline{i})}) \in \mathcal{T}^2 \mid ((\beta+1)^{(\underline{j})}, \alpha^{(\underline{i})}) \in D\}.$$

Some points to keep in mind for $A = K_{\mathcal{P}}(D)$ are:

(i) We have $0^{(\underline{i})} = 0^{(\underline{j})}$ in D, but $0^{(i)} \neq 0^{(j)}$ in A.
(ii) Arcs in D originating at $\mathbf{0}$ do not yield arcs in A.
(iii) Arcs in D originating at $\boldsymbol{\infty}$ yield d arcs in A (this agrees with what is observed in Remark 7.6).

Applying $K_{\mathcal{P}}$ to the digraphs $D_{\mathcal{P}}$ in Figure 2 which, recall, are reflexive, yields the corresponding digraphs $A_{\mathcal{P}}$ in Figure 4. The graphs in Figure 4 are irreflexive except $A_{\mathcal{P}}(L_3)$ which, being transitive, has loops on the vertices $3^{(2)}, 4^{(2)}$, and $5^{(3)}$. Checking easily that $K_{\mathcal{P}}(\mathcal{C}^\infty) = \mathcal{T}$ the following is an immediate consequence of Definitions 3.1 and 7.1. We require $\mathcal{I}$ to be closed because of point 3. mentioned just above.

Fact 7.10. *If $\mathcal{I}$ is a closed family of irreducible intervals then $A_{\mathcal{P}}(\mathcal{I}) = K_{\mathcal{P}}(D_{\mathcal{P}}(\mathcal{I}))$.*

Observe that this construction is clearly invertible, though only if defined on spanned extensions A of $\mathcal{T}$ coming from closed families $\mathcal{I}$. Its inverse is essentially a generalisation of Koh's construction from [5]; the only difference is that Koh's construction would have loops on $\mathcal{T}$. As he considered only posets, this difference would be cosmetic for him, but it is significant for us.

The construction $K_{\mathcal{P}}$ is useful in proving results such as the following.

Proposition 7.11. *Let $A = A_{\mathcal{P}}(\mathcal{I})$. If $\mathcal{I}$ is closed then A is transitive.*

Proof. Assume that $\mathcal{I}$ is closed. Then by Proposition 3.6 we have that $D = D_{\mathcal{P}}(\mathcal{I})$ is transitive, and by Fact 7.10 that $A = K_{\mathcal{P}}(D)$. Now assume that $\alpha^{(i)} \to \beta^{(j)}$ and $\beta^{(j)} \to \gamma^{(k)}$ in A. Then $(\alpha+1)^{(i)} \to \beta^{(j)}$ and $(\beta+1)^{(j)} \to \gamma^{(k)}$ in D. As we always have $\beta^{(j)} \to (\beta+1)^{(j)}$, we get an $((\alpha+1)^{(i)}, \gamma^{(k)})$-path in D. By the transitivity of D, $(\alpha+1)^{(i)} \to \gamma^{(k)}$ in D, and so $\alpha^{(i)} \to \gamma^{(k)}$ in A, as needed for transitivity. □ □

Lemma 7.12. *Let L be a sublattice of $\mathcal{P}$, and $A = A_{\mathcal{P}}(L)$. Then L is subdirect if and only if A is irreflexive, which is true if and only if A is acyclic.*

Proof. We have from Lemma 5.1 that L is subdirect if and only if $D_{\mathcal{P}}(L)$ has no edges of the form $\alpha^{(\underline{i})} \to (\alpha-1)^{(\underline{i})}$. As $A = K_{\mathcal{P}}(D_{\mathcal{P}}(L))$, this is true if and only if A is irreflexive. As $A = A_{\mathcal{P}}(\mathcal{I})$ for a closed family $\mathcal{I}$, it is transitive, so this is true if and only if A is acyclic. □ □

Similar to what we did in Section 6 we may view $\mathcal{T}$ as a *tournament decomposition* of A rather than viewing A as a spanned extension of $\mathcal{T}$, and get the following.

Corollary 7.13. *For every digraph A such that $\mathbb{A}_d(A) \cong L$, and every decomposition of A into a family $\mathcal{T}$ of d tournaments, there is an embedding of L into $\mathcal{P}$ such that $A = A_{\mathcal{P}}(L)$.*

Specialising to the case that A is acyclic (and so also irreflexive) then adding cosmetic loops, this gives the following result complementing Theorem 2.8. For a digraph A let A^r be the reflexive digraph we get by adding loops to every vertex.

Corollary 7.14. *For every poset P such that $\mathcal{A}_{\mathrm{m}}(P) \cong L$ there is a subdirect embedding of L into a product of chains $\mathcal{P}$ such that $P = A^r_{\mathcal{P}}(L)$.*

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